Data-Driven Optimization for Inventory, Credit Risk, and Supply Chain Management in New A-One & Co.

Final Submission *for the BDM Capstone Project*

Submitted by

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1. Executive Summary and Title

This capstone project addresses three major operational challenges faced by *New A-One & Co.*, a wholesale bangle distributor based in Firozabad, India: inventory inefficiencies, credit payment risks, and supply chain disruptions.

Using six months of primary data from June to December 2024, the project employs a data-driven approach combining descriptive analytics (Excel), predictive modelling (Python), and visualization tools to identify performance bottlenecks.

Inventory analysis reveals that AL design bangles contribute over 44% of sales but frequently stock out, while AH designs are overstocked and slow moving. Credit analysis highlights delayed payments totalling ₹2.56 lakhs, with two customers accounting for over 80% of dues. Supplier evaluation shows that while NavJivan Glass Works dominates procurement of DL/ML designs, Super Glass Works offers better performance for premium designs.

Recommendations include adopting data-driven inventory restocking strategies, implementing a credit scoring system for buyers, and diversifying the supplier base. Additionally, digitization of inventory and payment records is proposed for ongoing performance tracking. This project demonstrates the potential for transforming traditional businesses through structured data analytics and practical business intelligence tools.

This report demonstrates how even small B2B businesses can benefit from systematic data analysis to drive operational efficiency, minimize financial risk, and ensure long-term profitability.

Link to Proof of Originality:

Data Analysis File link: Click Here

Dataset Link: Click Here

2. Explanation of Analysis Procedure

In this analysis, a combination of advanced methods has been deployed, each of which contributes to addressing the challenges faced by New A-One & Co. .

i. Data Entry, Cleaning, and Preparation:

Physical records from New-A-One & Co.—including handwritten ledgers, invoices, and payment logs—were manually transcribed into Excel. The datasets covered 45 sales transactions, 40 supplier entries, and 9 customer payment profiles spanning June to December 2024. Basic cleaning steps involved standardizing dates, unifying naming conventions, and calculating derived fields like "Amount Left". The cleaned datasets were then imported into Python for structured analysis.

ii. Tools Used:

- MS Excel: Used for manual data entry, preliminary validations, and generating pivot tables for monthly summaries.
- Python: Employed for deep analysis, using pandas for data manipulation, numpy for numerical operations, and scikit-learn for modeling customer risk. Also used libraries like matpotlib.pyplot and seaborn for visualisation.

iii. **Analytical Models and Visualizations:**

- ABC Inventory Analysis: Used to classify bangle designs based on contribution to total sale helping prioritize stock replenishment and reduce wastage.
    - A (AL/DL): 20% SKUs, 80% revenue.
    - B (ML): Moderate movers.
    - C (AH/AM): Slow movers.
- **Credit Risk Assessment**: Customer payment behaviour was quantified using metrics such as overdue amount and frequency. High-risk customers were flagged based on poor compliance and frequent credit defaults.
    - Durga bangle store has best credit score of 6 followed by shariq bangle store with a credit score of 4

- Owais Bangle Store has a credit score of -1 indicating no payment has been done

Credit Risk	**Credit Terms**	**Actions**
Low (Score 6-7)	45-day terms + 2% early payment discount	Upsell premium designs (AL)
Medium (Score 3-5)	15-day terms + COD for high amount orders.	Weekly payment reminders via WhatsApp
High (Score ≤2)	Prepayment only	Blacklist after 2 defaults (like Owais)

- **Supplier Scorecard**: Created a score card that calculates total score based on timely deliveries, cost effectiveness and quality pass rate. Also created a 2D scatter plot (*Supplier Performance Analysis*) comparing supplier reliability (delivery time) with cost. This model highlighted which vendors delivered timely, good-quality, and cost-effective products.

Metric	**Weight**	**NavJivan**	**Super**	**Laxmi**
On Time Delivery	40	65	100	50
Cost Efficiency	30	95	85	70
Quality Pass Rate	30	90	98	88
Total Score		79	93	68

- **Time-Series Visualization**: Monthly sales and inventory data were plotted to detect patterns, such as seasonal spikes and overstock-stockout imbalances.

iv. **Customer Segmentation using K-Means Clustering:**

To better understand customer behaviour and payment trends, K-Means Clustering was applied to segment customers based on three key variables: total purchase amount, frequency of orders, and payment compliance ratio. This unsupervised machine learning technique partitions customers into distinct groups (clusters) such that customers within the same cluster exhibit similar purchasing and payment characteristics. For instance, one cluster revealed customers with high volume but poor payment discipline ideal candidates for stricter credit terms. Another cluster included low-volume but high compliant customers, ideal for upselling or loyalty incentives. This analysis enables targeted credit management, marketing strategies, and relationship building based on data-driven insights rather than intuition alone.

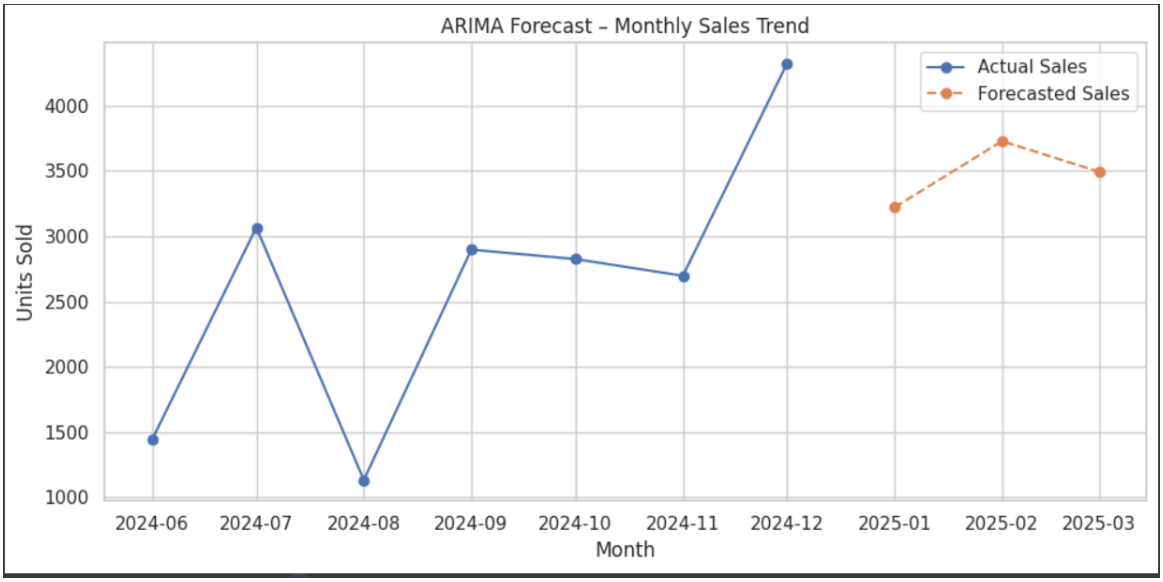
v. Business Insight Integration:

Qualitative inputs from the business owner were considered throughout the process. For example, months with high AL design sales matched with festive periods confirmed the need for demand-driven stock planning. Practical limitations like staff capacity and manual billing were also factored into the solution design, ensuring recommendations were both data-informed and feasible for the business to implement.

vi ARIMA Sales Forecasting

To predict future demand, an ARIMA (1,1,1) model was applied to monthly sales data from June to December 2024. The model captured seasonal trends and forecasted sales for the next three months, showing continued growth. This helps the

business plan inventory for high-demand designs like AL and avoid overstocking slow-moving items.



ARIMA Forecast of monthly sales to predict future sales

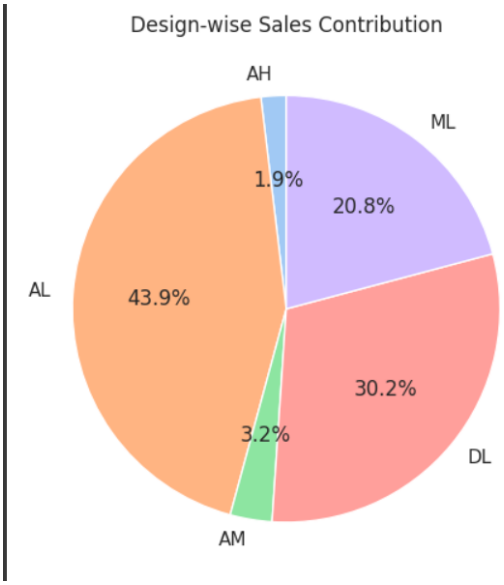
3. Results and Findings

The above-explained analysis methods were applied on the preprocessed data using Python programming language and its libraries such as Pandas, Numpy, statsmodel. The visualizations that are going to be presented here were created using Matplotlib, Seaborn. These visualizations will help us understand underlying patterns and trends, helping us to solve these problems.

i. Sales Performance:

Analysis of 45 transactions from June to December 2024 shows that total units sold were 18,384, with the AL design alone contributing 44% of sales volume and generating the highest revenue. DL and ML designs followed but with lower revenue-per-unit performance. This trend reinforces AL's position as the flagship product of New-A-One & Co.

Monthly sales peaked in December (4,322 units / ₹7.3 lakhs), coinciding with festival and wedding seasons, while the lowest occurred in June (1,444 units / ₹2.4 lakhs). The progressive growth across months (+120%) confirms strong seasonal demand patterns, which should influence stocking and promotional strategies.

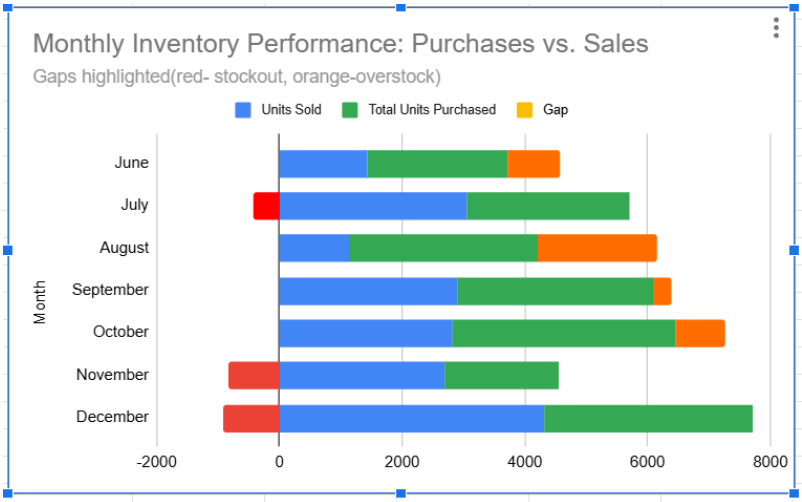


Sales Contribution by Designs

ii. Inventory Management:

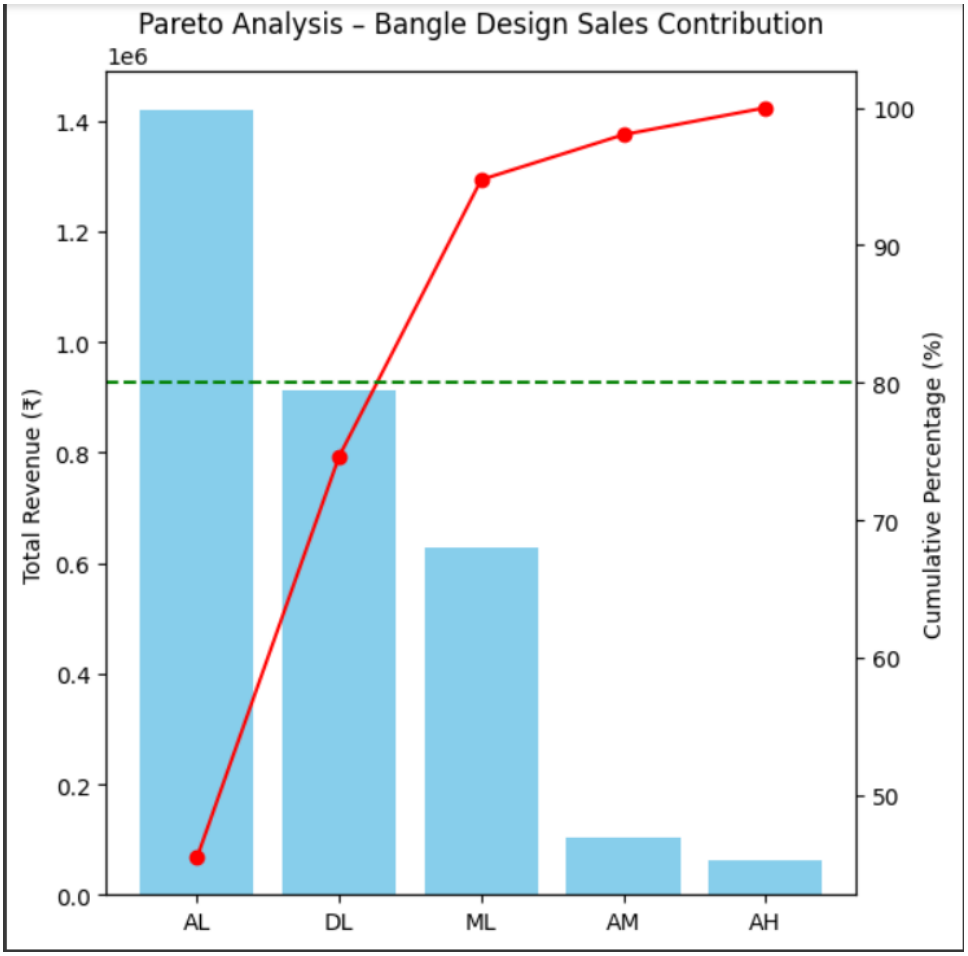
The inventory analysis revealed key inefficiencies:

- AL designs, despite being the most demanded, faced stockouts in July and November, causing lost sales.
- ML designs were overstocked during August and September, with units exceeding demand by up to 30%, tying up capital and storage space.
- Stocking decisions were not guided by past demand or sales cycles.



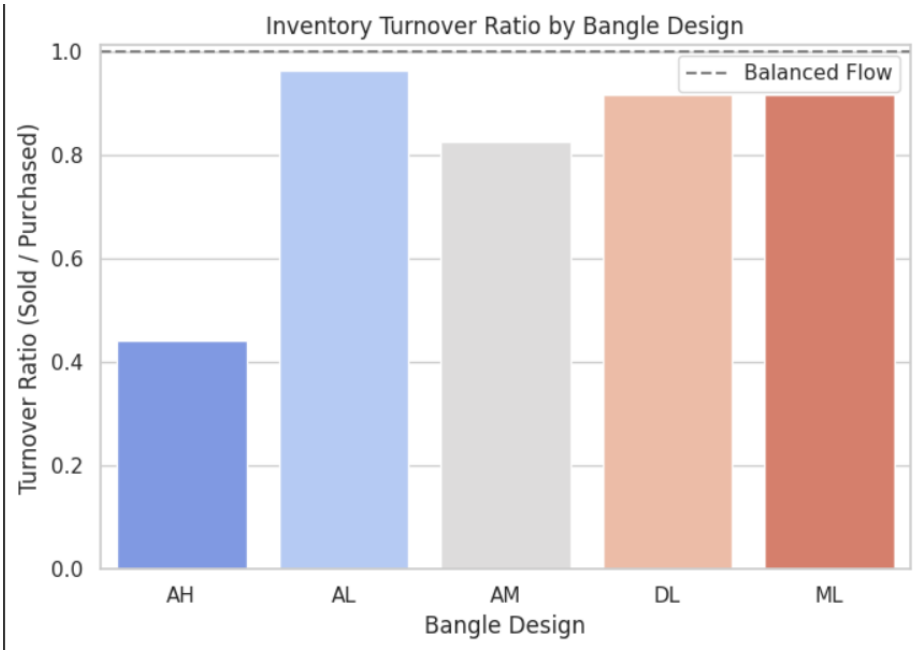
Monthly inventory Performance(Blue-Sold ,Green – Purchased, Orange-Overstock, Red-Stockout)

A Pareto analysis confirmed that AL and DL together generated over 80% of revenue, yet procurement was not prioritized accordingly. To support decision-making, an ARIMA model(*ARIMA Forecast of monthly sales to predict future sales)* was applied to the monthly sales data. The forecast suggested a continued upward trend in sales into Q1 2025, emphasizing the need to plan proactive restocking of fast-moving items like AL and DL in advance of peak seasons.



Pareto Analysis

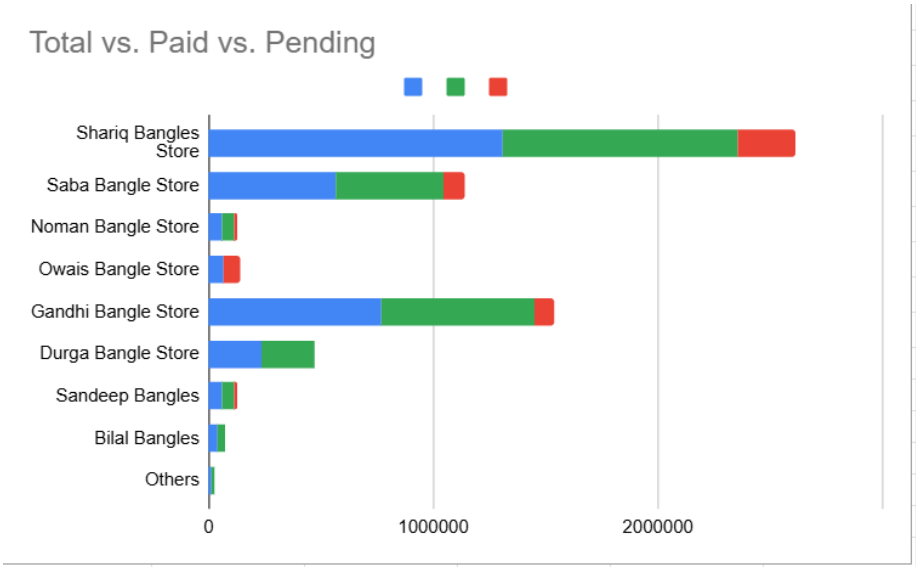
To evaluate how efficiently each bangle design was selling relative to its stock levels, an inventory turnover analysis was performed. This metric was calculated as the ratio of total units sold to total units purchased for each design between June and December 2024.



Inventory Turnover Ratios

Surprisingly, inventory turnover analysis revealed that even high-demand designs like AL and DL had turnover ratios below 1.0, indicating over-purchasing relative to actual sales. This suggests that the business may be stocking more units than necessary, even for fast-moving products. If this continues, it can lead to liquidity crunches and inefficient use of shelf space. The business should align procurement more closely with actual monthly sales patterns and consider implementing minimum-maximum thresholds for each design to balance stock flow.

iii. Payment and Credit Risk:



Credit Profile Analysis (Blue-Total, Green-Paid, Red-Due)

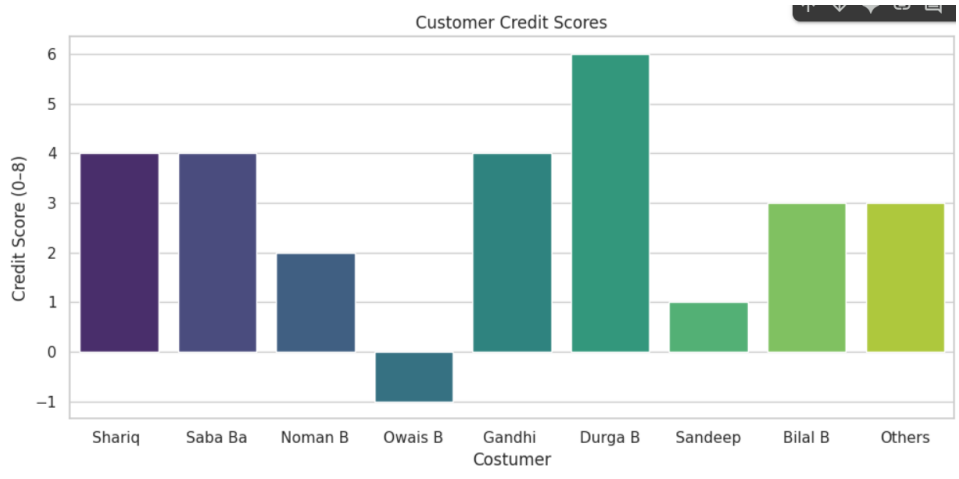
The credit profile analysis of 9 customers revealed that ₹2.56 lakhs was outstanding as of December 2024 20% of total billed revenue. Two customers, Shariq (₹2.56L) and Owais (₹68,992) accounted for huge overdue payments, representing significant credit risk.

Interestingly, customers with high purchase frequency showed lower payment compliance. For ex., Shariq made 16 purchases but maintained only 80.4% compliance, whereas Durga Bangle Store had perfect payment history with just 4 purchases.

A rule-based credit scoring model was developed using compliance %, total dues, and frequency. Based on this, customers were grouped into:

- Low-risk (Score 6–7) – Eligible for credit and loyalty offers
- Medium-risk (Score 3–5) – Credit allowed with tracking
- High-risk (Score ≤2) – Credit restriction or prepayment required

Durga Bangle Store showed the best credit score of 6 indicating a low risk costumer while Shariq bangle store despite being the top buyer showed the score of 4 only. Owais Bangle Store turned out to be the high risk costumer with a score of less than 2.



Costumer Credit Score

Customer Segmentation with K-Means Clustering:

A K-Means clustering model was used to classify customers into 3 groups based on total purchases, frequency, and payment compliance. The clusters provided deeper behavioural insights:

- Cluster 0 - High volume, low compliance ex. Shariq - needs stricter credit controls.
- Cluster 1 - Moderate risk ex. Gandhi and Saba - require monitoring.
- Cluster 2 - Low volume, high compliance ex. Durga - best candidates for upselling and loyalty programs.

This segmentation allows the business to customize credit terms, prioritize follow-ups, and build long-term relationships based on customer behaviour rather than intuition.



K Means Clustering of Costumers

iv. Supplier Insights:

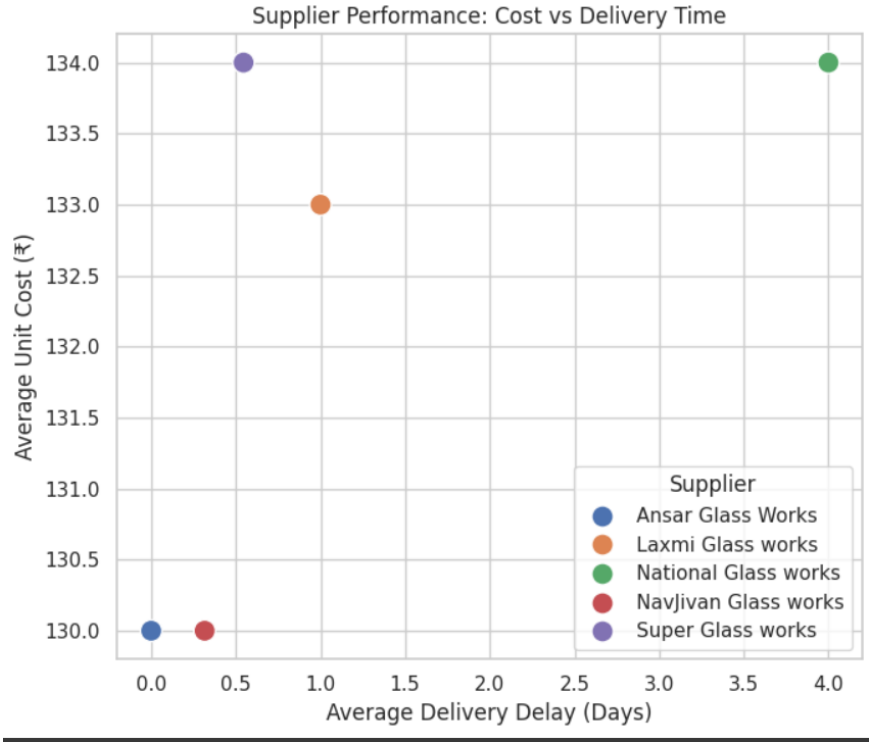
The supplier performance analysis was based on **40 procurement entries** from June to December 2024.

Key findings:

- NavJivan Glass Works handled over 60% of DL and ML procurement amd 45% of total units supplied at ₹130/unit but had longer delivery times, affecting order fulfillment.

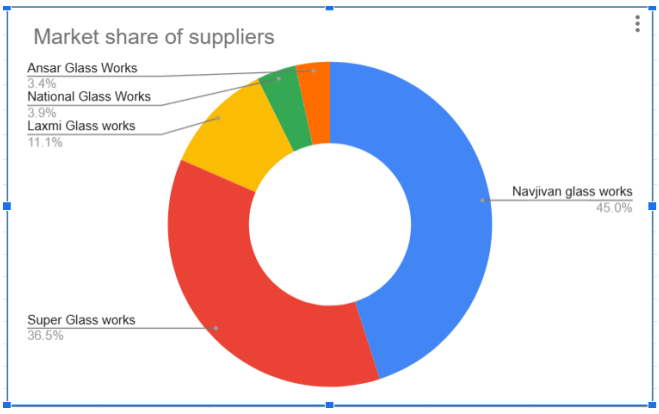
- Super Glass Works supplied AL designs at ₹134/unit slightly costlier, but with on-time delivery and high quality, making it the most reliable supplier.
- Laxmi Glass Works showed inconsistent delivery times and order quantities, signaling a high-risk vendor.

A 2D scatter plot of cost vs. delivery delay helped visualize each supplier's trade-off between cost efficiency and reliability. This insight is crucial to optimizing the vendor mix and negotiating contracts based on performance data rather than price alone



Supplier Performance Analysis

A supplier market share analysis revealed that NavJivan Glass Works accounted for the majority of purchases, contributing over 45% of total units supplied. While this ensures consistency, it also poses a single-supplier dependency risk. Super Glass works despite being a bit costlier turns out to be the most reliable supplier.



Supplier Market share of units

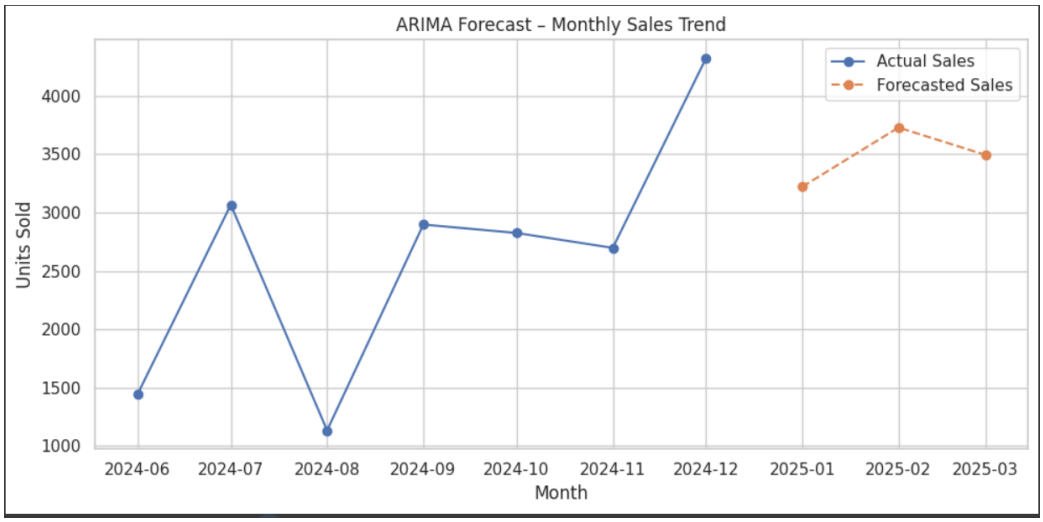
v. ARIMA Sales Forecasting

To predict upcoming demand, an ARIMA model with parameters (1,1,1) was applied to monthly sales data from June to December 2024.

Parameters Used:

- p=1 - it captures short-term sales momentum.
- d=1 – it ensures the data trend is stable.
- q=1 - it accounts for past forecasting errors.

The forecast predicted a continued but variable growth trend: 3,221 units in January, 3,729 units in February, and 3,494 units in March 2025. This indicates strong demand in Q1 with slight month-to-month fluctuations, reinforcing the need for stocking of high-demand designs like AL and DL. The forecast helps shift procurement decisions from reactive to predictive, enabling New-A-One & Co. to avoid stockouts and maintain optimal inventory levels during peak periods.



Monthly Sales Trends

Month	
2025-01-01	3220.665577
2025-02-01	3728.844082
2025-03-01	3494.359998

Forcasted Table

4. Interpretation of Results and Recommendations

This section interprets the analytical outcomes and translates them into clear, actionable strategies for New-A-One & Co. The recommendations are rooted in business logic and informed by data-driven insights derived from inventory trends, customer payment behaviour, supplier performance, and sales forecasting.

i. Sales Performance & Inventory Optimisation

Interpretation:

The inventory analysis highlighted a mismatch between procurement and actual product demand. Despite AL and DL being the best-selling designs, turnover ratios showed excess procurement relative to sales. Conversely, low-performing designs like ML and AH consumed inventory space and capital without proportional returns.

Strategies to implement:

An ABC classification was performed to group inventory based on contribution to total revenue:

- A-class: AL, DL - 80% of revenue, require tight stock control and priority procurement
- B-class: ML - moderate contribution, monitor monthly
- C-class: AH and AM - low contribution, avoid bulk purchasing

Recommendations:

- Implement a reorder point system based on historical monthly sales averages.
- Use inventory turnover analysis and Pareto classification to prioritize procurement of high-performing designs.
- Launch stock clearance discounts for underperforming designs (like ML, AH).

- Safety Stock = (Max Daily Sales × Max Lead Time) – (Avg Daily Sales × Avg Lead Time)
- Introduce design-wise minimum-maximum stock levels to avoid both shortages and excess.
- Use ABC + Inventory Turnover to structure stock reviews:
    - A-class: Weekly review, JIT (Just-In-Time) procurement
    - B-class: Biweekly stock check, moderate order quantity
    - C-class: Monthly review, only replenished upon sale

ii. Credit Risk and Costumer Management

Interpretation:

Shariq and Owais accounted for nearly 80% compliance. Despite frequent orders, they exhibited poor payment compliance. A custom credit score model and K-Means clustering exposed high-risk behaviour and helped group customers based on financial reliability.

Recommendation:

- Use the credit scoring model as a standard tool to assess all future customers.
    - Score 6–7 (Low Risk): Offer discounts, volume incentives
    - Score 3–5 (Moderate Risk): Require 50% upfront payment.
    - Score ≤2 (High Risk): Enforce full prepayment or cash on delivery
- Maintain a credit dashboard showing dues, risk flags, and contact logs.
- Add payment reminders via WhatsApp or SMS for customers.
- Monitor outstanding dues monthly using a visual dashboard or tracker.

iii. Supplier Evaluation and Risk Management

Interpretation:

NavJivan Glass Works, while cost-effective, has inconsistent delivery timelines, creating procurement delays. Super Glass Works, though pricier, provides timely and quality supply especially of AL designs. Supplier market share analysis revealed dependency on NavJivan posing operational risk. Laxmi Glass Works underperformed across all parameters.

Strategies to implement:

A Supplier Score Card was made that calculate Total Score of a Supplier based on

On timely deliveries, Cost Effectiveness and Quality pass rate.
NavJivan despite being the largest supplied scored a total of 79 on the other hand Super Glass a bit showed a total score of 93 despite being a bit costlier

Recommendations:

- Retain Super Glass Works as the primary AL design supplier due to reliability.
- Negotiate SLAs and delivery penalties with NavJivan to ensure timeliness.
- Limit orders to Laxmi Glass Works unless delivery consistency improves
- Identify and onboard at least one backup suppliers
- Use Score Card model to identify performance of Suppliers.

iv. Demand Forecasting with ARIMA

Interpretation:

The ARIMA(1,1,1) model forecasted monthly sales of 3,221 in January, 3,729 in February, and 3,494 in March 2025. While it confirmed steady growth, the fluctuations in February and March also suggest seasonality and potential mid-quarter dips. This enables strategic planning instead of reactive restocking.

Strategies to implement:

- Build a rolling 3-month forecast using ARIMA or Excel trend functions
- Link forecasts to reorder levels automatically (like projected units × 1.2 = reorder quantity)
- Use forecasted demand to negotiate early procurement discounts from suppliers during slow months
- Pair forecasting with ABC analysis to focus restocking on A-class designs
- Reorder Point Formula = **(Avg Daily Sales × Lead Time) + Safety Stock**

Recommendations:

- Base monthly purchase orders on ARIMA forecasts to prevent understocking.

- Maintain buffer stock of AL and DL designs during projected demand spikes.
- Use ARIMA monthly to continuously update forecasts and guide stocking decisions.
- Extend the forecast horizon to 6–12 months as more data becomes available.

v. Strategic Digitization and Reporting

Interpretation:

All data used in the analysis was manually extracted and digitized from handwritten records. This process is labour intensive, error-prone, and unsustainable as the business scales. Manual recordkeeping caused delays, and real-time decisions were impossible. The business currently operates reactively, with no data-driven tools for forecasting, credit control, or inventory tracking Decision-making currently lacks real-time insights.

Recommendations:

- Adopt Tally, Zoho Inventory, or free tools like Google Sheets POS templates.
- Schedule a monthly performance review using visual summaries: sales, dues, inventory turnover.
- Develop a supplier scorecard using:
  - Total units sold
  - Revenue vs forecast
  - Inventory by ABC class
  - Outstanding dues by customer
- Train 1 staff member on Excel formulas and data entry standard

5. Conclusion

This BDM capstone project has provided a structured, data-driven approach to addressing core operational challenges faced by New A-One & Co., a traditional B2B bangle wholesaler. Through manual digitization and analysis of sales, supplier, and customer data, key inefficiencies were uncovered in inventory management, credit risk handling, and procurement strategy.

Sales data analysis revealed that AL and DL bangle designs dominate business revenue, yet were either overstocked or frequently out of stock, highlighting a misalignment between procurement and real-time demand. Inventory turnover ratios and ABC classification provided a systematic framework for prioritizing restocking decisions. ARIMA-based time series forecasting further enabled demand projection for the first quarter of 2025, empowering the business to shift from reactive to predictive inventory management.

On the financial side, a credit scoring model and customer segmentation using K-Means clustering identified high-risk customers like Shariq and Owais, whose delayed payments made up over 80% of outstanding dues. By translating customer behaviour into quantifiable scores, the business is now better equipped to enforce structured credit policies and reduce financial exposure.

Supplier analysis using cost-delivery scatter plots and market share breakdowns revealed dependency on a few key vendors. Performance-based evaluation metrics were proposed to diversify risk and improve supply reliability.

Beyond analytics, the project emphasized the importance of digitization and low-cost automation through tools like Excel, dashboards, and POS systems. These recommendations offer not only immediate operational improvements but also lay the foundation for future scalability and resilience.

In conclusion, this project has demonstrated how small businesses can use basic data analytics to enhance decision-making, optimize processes, and build a more sustainable and intelligent operation—turning insights into real impact.